

Project plan

Project name: Estonian forest tree species classification from Sentinel-2 time series

Project members: Roles overlap where needed, the team meets once a week.

- Ruslan Nechshadimov: project manager (planning, deadlines, contact with the partner and the course staff), project report and presentations, web map for visualising labels and predictions.
- Kaaren Tenson: data preprocessing and model training.
- Juhan Pauklin: data preprocessing or modelling, to be confirmed after the first look at the data.
- Ander Saarniit: data preprocessing, model training, CI/CD.

Problem statement: Information about forest tree species is needed for forest management and monitoring, but field inventories are slow, expensive and get outdated because of felling and natural changes. Satellite images are collected regularly over the whole country, so they could help to keep species information up to date. The project predicts the dominant tree species of forest stands in Estonia from Sentinel-2 time series, using the Estonian Forest Register as training labels. It continues the work of a University of Tartu master's thesis (Liis Lille, 2026) which compared Random Forest and XGBoost in Western Estonia, and tests how well such a model works on held-out forest areas.

Objectives: The expected outcome is a reproducible pipeline and a species classifier for forest stands in one Estonian region, evaluated on held-out areas and compared with a baseline.

- Baseline: Random Forest on Sentinel-2 time series features. For reference, the partner's thesis reports overall accuracy about 0.65 and macro-F1 about 0.53 for a simple single-level Random Forest.
- Target: our best model beats the baseline on the same held-out areas. **To be decided by the team** (overall accuracy of 0.80 or more and macro-F1 of 0.60 or more. The thesis reaches 0.87 and 0.69 with a hierarchical XGBoost on the same data. Because we use the same data and split, our results can be compared with the thesis directly)
- **To be decided by the team** (show results per species and which time periods and features matter most)
- **To be decided by the team** (extension: one of a neural network on the time series, additional features, or Sentinel-1)
- A simple web map where stands, registry labels, predictions and errors can be inspected.

Data: The partner (Liis Lille) has given us the input data for Subregion 2 (Western Estonia), year 2025, the same files that her thesis models were trained on. The Forest Register and Sentinel-2 are also public, so we can download data for another region ourselves (the team has a Copernicus Data Space account).

- Labels: Forest Register stand polygons with the dominant species (Estonian Environment Agency), 9,776 stands in the modelling split. Classes: pine, spruce, birch, grey alder, black alder, aspen, other.

- Features: ready Sentinel-2 L2A tables from the Sentinel Hub Statistical API, one row per stand and 10-day period in 2025 (10 spectral bands and 4 vegetation indices, per-stand mean). Train 6,909, validation 1,252 and test 1,615 stands.
- Extra: per-stand height statistics (Environment Agency height raster) and the majority soil code (Land Board soil map) in one table, a list of 659 stands that overlap clear-cuts and the spatial split from the thesis.

Main steps:

1. Load the tables and pivot the time series to one row per stand.
2. Remove the stands that overlap clear-cuts and check the class distribution and missing values.
3. Use the provided spatial split, so our results are comparable with the thesis.
4. Fill missing values (imputation fitted on the training set only). **To be decided by the team** (optionally add the height and soil features)
5. Run the baseline and the other models on the same tables.
6. **To be decided by the team** (if we test another region: download stand polygons from the public Forest Register service and Sentinel-2 statistics for them, on the same 10 m grid (EPSG:3301))

Methodology: Supervised multi-class classification of forest stands using features from the Sentinel-2 time series.

- Baseline: Random Forest (scikit-learn) with balanced class weights on all Sentinel-2 features.
- Gradient boosting: XGBoost, hyperparameters tuned on the validation set.
- **To be decided by the team** (hierarchical variant as in the partner's thesis: conifers, broadleaves and other first, then species inside each group)
- **To be decided by the team** (our own extension, chosen after the baseline works: a neural network on the time series (1D-CNN or LSTM), additional phenology-based features, or a test in a second region. Sentinel-1 and LiDAR only if time remains)
- **To be decided by the team** (class imbalance is handled with class weights and oversampling of rare classes in the training data only)
- The partner's open code is used as a starting point and cited.

Evaluation: Models are compared on stands from grid cells that were never used for training or tuning.

- Metrics: overall accuracy, macro-F1 (main metric because classes are imbalanced), confusion matrices. **To be decided by the team** (possibly also balanced accuracy, per-class precision, recall and F1, Cohen's kappa)
- Validation set is used for tuning, the test set only once for the final models.
- All models are compared with the baseline on the identical split. **To be decided by the team** (repeat with spatial cross-validation to see the variation of results)
- The class 'other' has very few stands, so macro-F1 is also reported for the main six classes.

Expected challenges:

- Spatial leakage: neighbouring stands are similar and can make results too optimistic. Addressed by splitting by 5 x 5 km grid cells.
- Class imbalance: pine and birch make up most stands, aspen and 'other' are rare. Addressed with class weights, oversampling in the training set and macro-F1.
- Clouds, snow and gaps in the time series: addressed with the scene classification mask and 10-day windows, winter windows without data are dropped.
- Noisy or outdated labels: the registry can be older than the images and stands may have been felled. Addressed with the provided clear-cut list and visual checks of suspicious stands on a map.
- Data volume and service limits: start with one region and one year.
- **To be decided by the team** (mixed stands: start with stands where one species has at least 80%, mixed stands later)
- Aligning data layers: all sources must use the same grid and resolution, planned from the start.

Resources and tools:

- Python: geopandas, rasterio, pandas, scikit-learn, XGBoost, Jupyter notebooks. **To be decided by the team** (PyTorch)
- Copernicus Data Space Ecosystem account with OAuth keys (already created), Sentinel Hub Statistical API.
- Git repository with CI (the partner's data is not committed, because the Forest Register extract is licensed). **To be decided by the team** (QGIS for checking data)
- JavaScript web map (MapLibre or Leaflet) for visualisation.
- Local computers should be enough for one region. **To be decided by the team** (if not, the University of Tartu HPC, to be checked)

Questions for further guidance: **To be decided by the team**

1. Is one region and one year enough, or should we also test on a second region?
2. Can we use the partner's data for testing in another subregion, and where can we get the clear-cut layer for it?
3. How much of the partner's code may we reuse and what counts as our own contribution?
4. Is Sentinel-1 or LiDAR expected, or optional if we have time?
5. Is the thesis setup a good enough baseline, or should we also add a simpler one, for example majority class or logistic regression?

Milestones and timeline: **To be decided by the team**

WEEK	DATE <i>(the marked date is the end date of the running week)</i>	MILESTONES OR PLANNED ACTIVITIES
Week 3	September 28	• Topic selection.

Week 4	October 4 <i>(teams formed, project chosen)</i>	• Team formed and project chosen. • Partner's thesis and code read, roles discussed.
Week 5	October 11	• Look at the partner's data, load and pivot the tables. • Set up the repository and environment (data stays out of git). • Draft the project plan.
Week 6	October 18 <i>(project plan deadline)</i>	• Finish and submit the project plan. • Data exploration, remove clear-cut stands. • Baseline Random Forest on the provided split.
Week 7	October 25	• Baseline results and exploration plots. • XGBoost.
Week 8	November 1	• Hierarchical model. • Choose our own extension.
Week 9	November 8	• First version of the extension (neural network, new features or another region). • Start the web map.
Week 10	November 15	• Prepare the intermediate presentation. • Continue the extension.
Week 11	November 22 <i>(intermediate presentations during this week Nov 16-18)</i>	• Intermediate presentation. • Use the feedback, continue the extension.
Week 12	November 29	• Extension results and hyperparameter tuning. • Web map with predictions.
Week 13	December 6	• Error analysis. • Optional: Sentinel-1.
Week 14	December 13	• Freeze final experiments. • Prepare the final presentation, write the report.
Week 15	December 20 <i>(final presentations and team evaluation during this week Dec 14-16)</i>	• Final presentation. • Finish the report.